PARISHRAM 2.0

CLASS 12TH 2026

Inverse Trigonometric Functions

MATHS **LECTURE-1**

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TOPICS TO BE COVERED

1. Trigonometric Functions (RECAP)
2. Inverse Trigo Functions
3. Principal Value

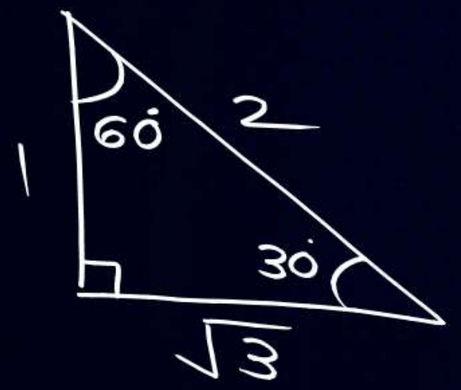
Concept

Trigo Functions

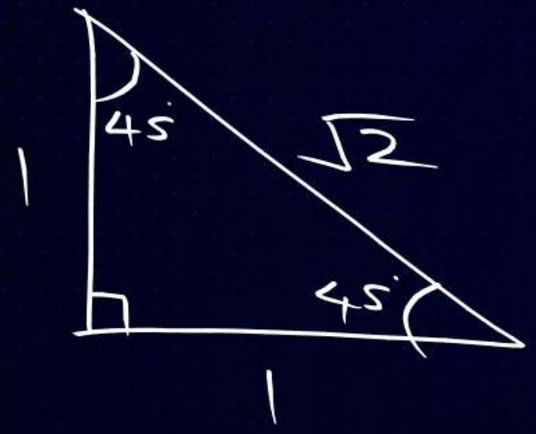
$\sin 0^\circ = 0$
$\sin 90^\circ = 1$
$\cos 0^\circ = 1$
$\cos 90^\circ = 0$

Trigo Values

30-60-90



45-45-90



$\sin 60^\circ = \frac{O}{H} = \frac{\sqrt{3}}{2}$

$\cos 45^\circ = \frac{A}{H} = \frac{1}{\sqrt{2}}$

$\sin 30^\circ = \frac{O}{H} = \frac{1}{2}$

$\sin 45^\circ = \frac{O}{H} = \frac{1}{\sqrt{2}}$

$\tan 45^\circ = \frac{O}{A} = \frac{1}{1}$

Inverse Trigonometric Functions



$f(x) = \sin x$

$\sin 45^\circ = \frac{1}{\sqrt{2}}$

Trigo Fⁿ

$45^\circ = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$

Inverse Trigo

$\sin^{-1} x = y$

$\tan 45^\circ = 1 \rightarrow$ Trigo Function

$45^\circ = \tan^{-1}(1)$

Inverse Trigo Fⁿ

Angle → Degree
→ Radian

Sirf ★★★ $\boxed{180^\circ = \pi^c}$

$30^\circ = x^c$

$180 \times x = 30 \times \pi$

$x = \frac{30 \times \pi}{180}$

$x = \frac{\pi}{6}^c$

$30^\circ \rightarrow \frac{\pi}{6}$

$180 = \pi^c$

$45^\circ = x^c$

$x = ?$

a) $\pi/6$

b) $\pi/4$

c) $\pi/3$

d) $\pi/8$

$\frac{\pi}{4} = \frac{180}{4}$

$\rightarrow 45^\circ$

Inverse Trigonometric Functions

$\sin^{-1}x$
$\cot^{-1}x$
$\tan^{-1}x$

$\text{cosec}^{-1}x$
$\sec^{-1}x$
$\cot^{-1}x$

$$\sin 45° = \frac{1}{\sqrt{2}}$$

$$45° = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$\frac{\pi}{4} = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$\sin 30° = \frac{1}{2}$$

$$30° = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\frac{\pi}{6} = \sin^{-1}\left(\frac{1}{2}\right)$$

Functions



$\sin 45 = \frac{1}{\sqrt{2}}$ $\quad \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$

For the function to be invertible.
It should be one-one & onto

one-one

↳ Ek Ek se hi connected

Onto

Range = Codomain

Trigo Function

$\sin 60^\circ = \frac{\sqrt{3}}{2}$

Input $\rightarrow 60^\circ$

Output $\rightarrow \frac{\sqrt{3}}{2}$

Inverse Trigo

$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60^\circ$

Input $\rightarrow \frac{\sqrt{3}}{2}$

Output $\rightarrow 60^\circ$

Trigo Fn $\rightarrow$ Domain

$\rightarrow$ Range

Inverse $\rightarrow$ Domain

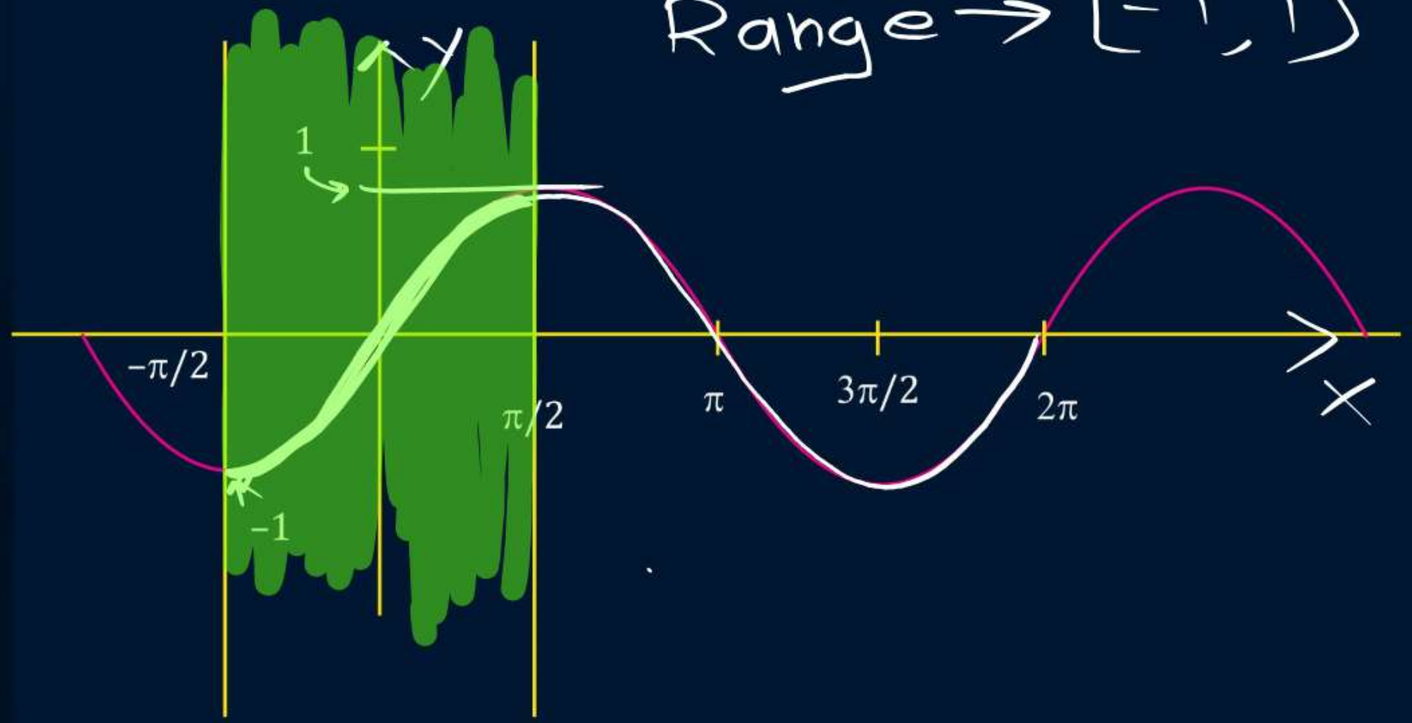
$\rightarrow$ Range

Inverse Trigonometric Functions

$y = \sin x$

Domain → $-\frac{\pi}{2}, \frac{\pi}{2}$

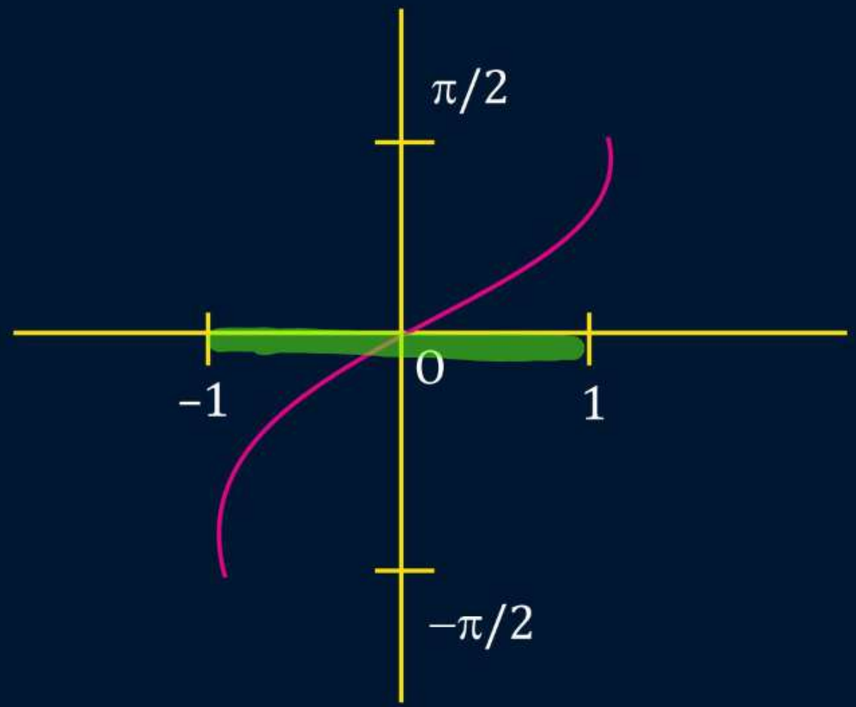
Range → $[-1, 1]$



$y = \sin^{-1} x$

Domain → $[-1, 1]$

Range → $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$



Domain and Range of Inverse Trigo Functions

Function	Domain	Range
$\sin^{-1}x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}x$	~~$(-\infty, \infty)$~~ $\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1}x$	~~$(-\infty, \infty)$~~ $\mathbb{R}$	$(0, \pi)$
$\sec^{-1}x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
$\operatorname{cosec}^{-1}x$	$\mathbb{R} - (-1, 1)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

Output

$\sin^{-1}(1) = \frac{\pi}{2}$

Input

Question

Find the principal value of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$.

$\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = x$

$\frac{1}{\sqrt{2}} = \sin x$

$\sin\frac{\pi}{4} = \sin x$

$x = \pi/4$

$x = 45°$

Range $\rightarrow \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

$[-90, 90°]$

Question

Find the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$.

$\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = x$

$\frac{\sqrt{3}}{2} = \cos x$

$\cos\frac{\pi}{6} = \cos x$

$x = \frac{\pi}{6}$

$[0, \pi] \rightarrow$ Range

THANK YOU